Reproducing *Finite-Sample Adequacy and Robustness of Null Distributions*

A vignette for the mc-tk simulation package (COST-D-26-00503R1, *Computational Statistics*)

The authors

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1. Purpose and scope

This vignette is the reproducibility document for the study. It explains what the code does, in which order to run it, which software versions were used, and how a reader can confirm in about one second that a downloaded copy behaves exactly as the copy that produced the manuscript. Every command whose output is shown below was executed on the machine described in Section 3, and the output is the output that command produced; the only unexecuted line in the document is the `cd` into the repository folder in Section 2, whose name depends on where the package is downloaded.

The study itself asks three questions: at what sample size the null distribution of each of fourteen tests becomes adequate (RQ1), how the null distribution degrades when assumptions are violated (RQ2), and how many Monte Carlo replications are needed to verify either claim reliably (RQ3); a fourth, companion study measures power so that calibration is not confused with performance. The table below maps each reproducibility requirement to the artefact that answers it.

Requirement	Where it is answered
“an additional vignette … in pdf format that is easily accessible”	this file, `vignette.pdf`
“all required dependencies … with exact version numbers”	Sections 3 and 10; `requirements.txt`, `requirements-lock.txt`, `docs/pip_freeze_full.txt`, `docs/ENVIRONMENT.txt`
“contents of each code file … should be explained”	Section 5; `docs/CODE_MAP.md`
“a flow chart … how and in which sequence the scripts should be run”	Section 4, Figure 1; Mermaid source in `docs/flowchart.mmd`
“guidance … how a non Python user could set up the required software environment with short illustrative examples”	Section 9
“a separate web page that only contains the code used in this study”	Section 11; `docs/REPO_LAYOUT.md`
“at least provide a short description to each function”	Section 7; docstrings in every source file; `docs/FUNCTION_REFERENCE.md`

What the package contains. Twenty-five Python files, 11,980 lines in total. Eight study scripts carry the analysis (`core.py`, the shared kernel, plus `rev_study1_ext.py`, `rev_sensitivity.py`, `rev_power.py`, `rev_dcor.py`, `rev_imbal_hetero.py`, `rev_alpha_dependence.py` and the frozen `archive_experiments.py`, together 9,352 lines); three generators produce the two long runs whose results are committed rather than re-derived (`gen_t12_law.py`, `gen_dcor_local.py`, `gen_t14_study3.py`); five tools verify the package (`verify_manuscript.py`, `verify_crossrefs.py`, `verify_letter_claims.py`, `verify_new_tests_power.py`, `verify_dcor_against_R.py`); two build the published figure and the editable manuscript (`make_fig2.py`, `post-process_docx.py`); and seven more sit in `docs/`. Alongside them are the results files in `results/`, the figure files in `figures/`, the LaTeX build (`build.sh`, `latex_preamble.tex`) and the documentation tooling. Everything in the paper is regenerated by these scripts; no number in the manuscript was entered by hand, and `verify_manuscript.py` checks that claim mechanically (Section 10).

2. Quick start

The commands below are the whole recipe. They assume the repository has been downloaded and that `python3` is available (Section 3 explains how to obtain it).

```
cd "Finite-Sample Adequacy and Robustness of Null Distributions"
python3 -m pip install -r requirements-lock.txt   # exact versions of Sec. 3
bash docs/run_verification.sh                     # all twelve read-only checks (~2 min)
python3 docs/verify_bartlett.py                   # reproduce one published number
python3 experiments.py                            # regenerate Studies 1–3 (~3 min)
```

`docs/run_verification.sh` runs the whole read-only check set and writes `docs/verification_log.txt`: the kernel self-test (20 checks), the numeric cross-checks against the committed results files, the worked example below, the R cross-check for the distance-correlation statistic, and the SHA-256 manifest verdict. It modifies nothing, so the log records the committed state rather than a rebuild. To regenerate the derived artefacts (the pinned lock, the manifest, the flowchart, the function reference) run `docs/regenerate.sh` first; that one is a build step, which is why it is kept out of the verification script. The last command regenerates the archive artefacts.

Each script writes its results file into `results/` (or beside itself in the submission layout), so running one overwrites the committed copy of its own results and nothing else. To leave the committed files untouched, run in a copy of the directory or pass `--outdir DIR`, which the analysis scripts accept and which creates the directory if it does not exist (for example `python3 code/rev_study1_ext.py --outdir /tmp/run --validate`). Section 10 gives

the per-study commands, expected runtimes and the key numbers to compare against.

One scope note. Most of the study re-runs from a clone in minutes, but three analyses are long enough that their results are committed as artefacts rather than regenerated by the verification script: the T12 reference law (about 110 minutes), and the T14 local-power and Study 3 precision grids (about four minutes each). `README.md` gives the cost of every analysis in one table, and Section 5 marks the three generators.

3. Computing environment

All results in the manuscript were produced with the following environment (captured from the live interpreter by `docs/make_lock.py`, never typed by hand):

Item	Value
Operating system	macOS 26.6.2 (Darwin, arm64, Apple Silicon)
CPU cores	10
Python	CPython 3.13.5, Anaconda distribution, `/opt/anaconda3/bin/python3`
pip	26.1.2
Direct dependencies	numpy 2.4.6, scipy 1.16.3, matplotlib 3.11.0, lightgbm 4.6.0
Toolkit referenced in Section 4 of the paper	mc-tk 1.1.0 (not imported by the simulation pipeline)

`requirements.txt` declares the four direct dependencies with lower bounds and the reason each is needed. `requirements-lock.txt` pins the full runtime closure — four direct dependencies plus their nine transitive dependencies — to the exact versions above:

```
contourpy==1.3.3        lightgbm==4.6.0         pyparsing==3.3.2
cycler==0.12.1          matplotlib==3.11.0      python-dateutil==2.9.0.post0
fonttools==4.63.0       numpy==2.4.6            scipy==1.16.3
kiwisolver==1.5.0       packaging==26.2         six==1.17.0
pillow==12.3.0
```

The complete freeze of the producing machine (721 entries) is in `docs/pip_freeze_full.txt`. It is a base Anaconda environment in which several unrelated projects are also installed, so the file records that 297 entries referring to private local paths were removed and why; none of them is imported by this study. Three packages that readers often expect here — `pandas`, `scikit-learn` and `statsmodels` — are installed but **not** used: every statistic in the paper is implemented directly with NumPy/SciPy, and the only machine-learning dependency is LightGBM for the interaction test T13.

Two facts about provenance that matter.

1. The results were produced by the standalone reproduction scripts, not by the packaged mc-tk library. Section 4 of the manuscript states this explicitly, and the package is consistent with it: **no script in this package imports `mc-tk`** (`grep -n "import mc" *.py` returns nothing), and the library only illustrates the object-oriented API in Section 4 of the paper.
2. The mc-tk package installed on the producing machine is version **1.1.0**, which is the version the manuscript cites. The toolkit is not needed to regenerate any result, so no result depends on which version is installed.

Consistency check of these pinned versions against the archived runs. Every results file records the versions of the interpreter and libraries that wrote it, so the pins can be checked against the actual runs rather than asserted:

Results file	numpy	scipy	python
`rev_results_t1.json` (`meta` and `meta_alpha_dep`)	2.4.6	1.16.3	3.13.5
`rev_results.json`	2.4.6	1.16.3	3.13.5
`rev_results_power.json`	2.4.6	1.16.3	3.13.5
`rev_results_newtests_power.json`	2.4.6	— (not imported)	3.13.5

All four agree with `requirements-lock.txt`. The only file without a `meta` block is the archive `experiment_results.json` (produced by the archive pipeline); its provenance is covered by the checksum manifest and by the full re-run recorded in Section 10, which reproduces every integer rate exactly under exactly these pinned versions. A clean virtual environment built from `requirements-lock.txt` alone runs the worked example successfully, which is the strongest available evidence that the lock file is complete rather than merely consistent.

Installing. Any Python 3.10+ installation with NumPy, SciPy and Matplotlib will run the pipeline; the exact versions are needed only for bit-level agreement. Section 9 gives the step-by-step version for readers who do not use Python.

4. Execution flowchart

Figure 1 shows the order in which the files are used: the environment is set up once; `experiments.py` holds the ten archived statistic generators and is imported unchanged by `core.py`, the shared kernel of the extended panel (whose `_selftest` runs inside every script); the kernel is then used by the four sensitivity and power study scripts and by the independent power check, each of which writes one or two result files; the result files are the only inputs to the tables and figures of the manuscript; and a verification layer checks the whole tree. One ordering constraint exists: `rev_study1_ext.py` must run before `rev_alpha_dependence.py`, because the second merges its blocks into the file the first created.

The same chart is available as Mermaid source (`docs/flowchart.mmd`), which GitHub renders automatically, and as a vector PDF (`docs/fig_execution_flow.pdf`) for inclusion in other documents. The source in condensed form:

```
flowchart TD
    ENV["Step 0: pip install -r requirements-lock.txt"]
    EXP["experiments.py — T1–T10 generators, Studies 1–3 (~3.3 min)"]
    CORE["core.py — registry TESTS (T1–T13), _evaluate(), _selftest()"]
    ENV --> EXP --> CORE
   CORE --> S1["rev_study1_ext.py (~38 min)"] --> O1["rev_results_t1.json, fig_rev_alpha.png"]
       S1 -.->|merges into the same file| S2["rev_alpha_dependence.py (~5 min)"] --
> O2["fig_rev_dependence.png"]
    CORE --> S3["rev_sensitivity.py (~1 min)"] --> O3["rev_results.json + 2 figures"]
   CORE --> S4["rev_power.py (~1 min)"] --> O4["rev_results_power.json + fig_rev_power.png"]
   CORE --> S5["verify_new_tests_power.py (37 s)"] --> O5["rev_results_newtests_power.json"]
    EXP --> O0["experiment_results.json + fig_study1..3.png"]
    O0 --> MAN["Tables and figures of the manuscript"]
    O1 --> MAN
    O2 --> MAN
    O3 --> MAN
    O4 --> MAN
    O5 --> MAN
   MAN --> VER["verify_manuscript.py (235 checks) · verify_bartlett.py (13/13) · make_manifest.py --
check"]
```

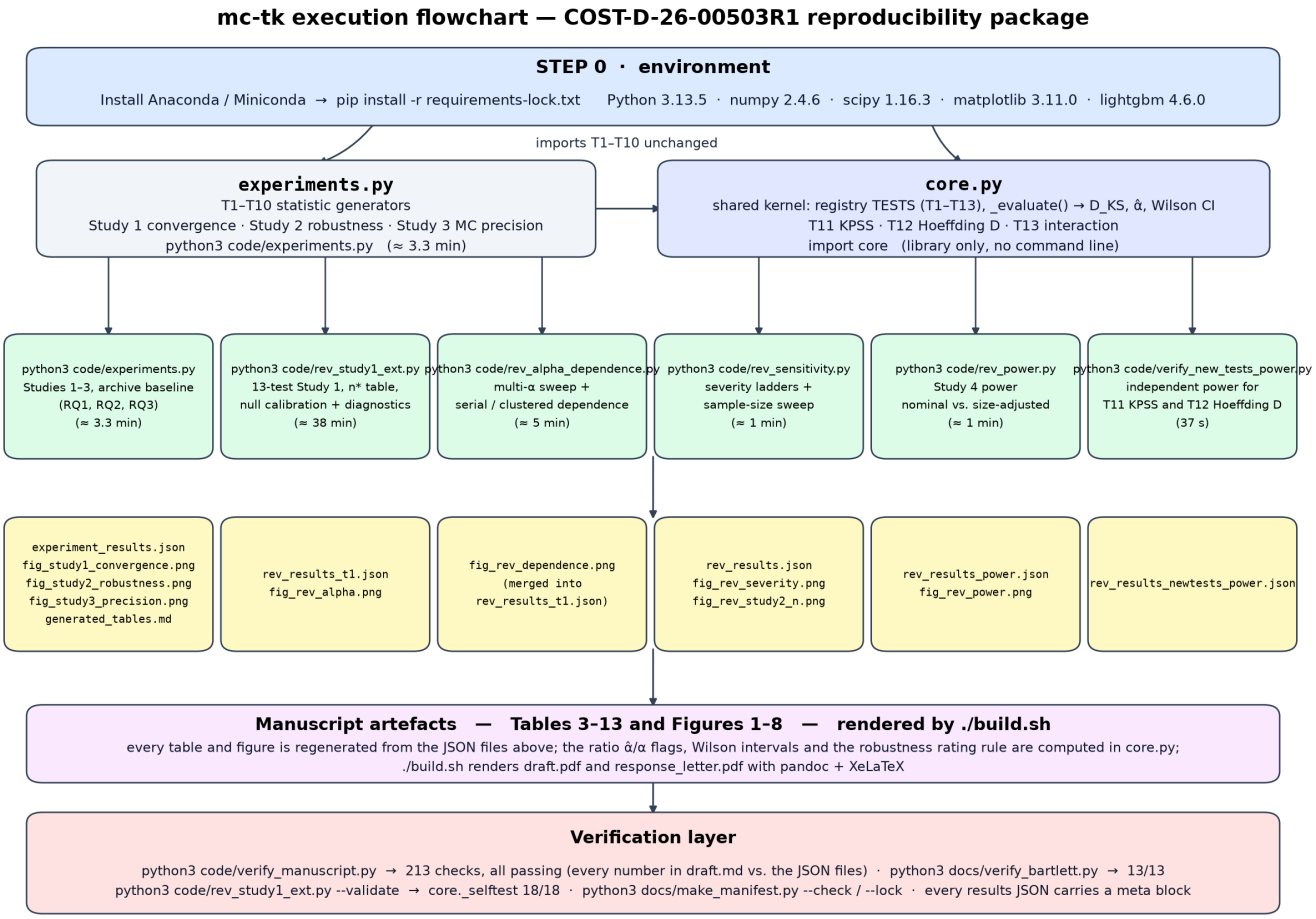


Figure 1: Execution flowchart of the mc-tk simulation package. The archive script and the shared kernel are at the top, the five sensitivity and verification scripts in the middle with their measured runtimes, their result files below them, and the verification layer at the bottom.

5. What each code file does

The full file-by-file description, with inputs, outputs, runtime and dependencies, is `docs/CODE_MAP.md`. In brief:

File	Purpose	Command	Runtime
`experiments.py`	Frozen original-submission script: generators T1–T10, divergence measures, Study 1 (convergence), Study 2 (robustness), Study 3 (MC precision), table generator. Byte-identical to the version published with the first submission and deliberately not edited	`python3 experiments.py`	169 s
`code/archive_experiments.py`	The same file, under the name `core.py` imports it by. Identical content; kept so that the package root copy stays an untouched record	imported	—
`code/core.py`	Shared kernel: the TESTS registry for the 14-test panel, the single evaluation path `_evaluate()` returning D_{KS}, $\widehat{\alpha}$ and the Wilson interval, generators for T11 (KPSS), T12 (Hoeffding's D), T13 (interaction difference) and T14 (distance correlation), the dependence generators, the two T12 reference objects, `_selftest()` (20 checks) and the results-file `meta` writer	`import core`	import, ~1 s
`code/rev_study1_ext.py`	Study 1 grid at three nominal levels, the n^* table, Monte-Carlo null calibration of T11–T13, the T12 reference diagnostic and the T13 split diagnostic, plus `fig_rev_alpha.png`	`python3 code/rev_study1_ext.py` (`--validate` runs the verification suite only)	≈ 38 min (2302 s)
`code/rev_dcor.py`	T14 convergence by dimension, the within-vector-correlation nuisance check, and power	`python3 code/rev_dcor.py`	1103 s

File	Purpose	Command	Runtime
`code/rev_alpha_dependence.py`	Study 2 scenarios at $\alpha \in \{0.01, 0.05, 0.10\}$ for the 13 tests evaluated there, the tail-mass identity checks, and the dependence designs (AR(1) serial, clustered/ICC, beta-binomial blocks, KPSS under stationary AR(1), T13 under correlated covariates)	`python3 code/rev_alpha_dependence.py`	≈ 5 min (288 s)
`code/rev_sensitivity.py`	Severity ladders for every violation type at $n = 50$, and the Study 2 sample-size sweep $n \in \{20, 50, 100, 200\}$	`python3 code/rev_sensitivity.py`	76 s
`code/rev_imbal_hetero.py`	The crossed allocation $\times$ variance design of Table 8, which no one-factor ladder can see	`python3 code/rev_imbal_hetero.py`	~3 min
`code/rev_power.py`	Study 4: nominal-α versus size-adjusted power, with equivalence and calibration self-checks	`python3 code/rev_power.py`	53 s
`code/gen_t12_law.py`	**Committed-artefact generator**: the T12 reference law (one realisation at $n_{\text{law}} = 2{,}000$, $N = 200{,}000$) and the diagnostic table. `--diag-only` reuses the committed grid and takes about 4 min	`python3 code/gen_t12_law.py`	≈ 110 min (6580 s)
`code/gen_dcor_local.py`	Generates the T14 local-power grid against a spatially concentrated signal (Supplementary §S10)	`python3 code/gen_dcor_local.py`	214 s
`code/gen_t14_study3.py`	Generates the T14 Study 3 precision curve, 30 independent runs	`python3 code/gen_t14_study3.py`	213 s
`code/verify_manuscript.py`	Cross-check of every number asserted in the manuscript against the results files that produced it; read-only. The text checks are skipped, not failed, when the manuscript is absent	`python3 code/verify_manuscript.py`	0.3 s
`code/verify_crossrefs.py`	Audits figures and tables: captions in order, every reference resolving, every figure cited. Skips when the manuscript is absent	`python3 code/verify_crossrefs.py`	0.1 s

File	Purpose	Command	Runtime
code/verify_letter_claims.py	Checks that every artefact the response letter claims to have produced exists on disk	python3 code/verify_letter_claims.py	0.1 s
code/verify_new_tests_power.py	Independent power measurement for the two new continuous tests (T11, T12)	python3 code/verify_new_tests_power.py	37 s
code/verify_dcor_against_R.py	Compares the T14 statistic against energy::dcorT in R; needs R with the energy package	python3 code/verify_dcor_against_R.py	~20 s
code/make_fig2.py	Regenerates figures/fig_study2_robustness.png with the legend outside the axes	python3 code/make_fig2.py	~2 s
code/postprocess_docx.py	Enables the Word table-of-contents refresh and fits the tables to the text column in the editable manuscript	python3 code/postprocess_docx.py draft.docx	~2 s
build.sh, latex_preamble.tex	Render the manuscript, the response letter and the supplement with pandoc + XeLaTeX; ./build.sh docx also produces the editable Word files. The documents it builds are submission material and are not distributed here, so this script is useful only in the submission copy	./build.sh	seconds
docs/run_verification.sh / docs/regenerate.sh	Read-only verification / derived-artefact regeneration. The split is deliberate: a verification run must not rewrite the environment record it is checking	bash docs/run_verification.sh	~2 min

The run order matters in one place: rev_study1_ext.py writes results/rev_results_t1.json and rev_alpha_dependence.py merges its blocks into that same file, so it must run second. All other scripts are order-independent. Every script, and every docs/ tool, is described in docs/CODE_MAP.md, and the function-level reference docs/FUNCTION_REFERENCE.md is generated from the sources by docs/make_function_reference.py.

6. Running the pipeline, step by step

Each step below states the command, the artefacts it writes, the measured runtime, and the key numbers that tell you the run succeeded. All commands are run from the package root (the directory containing draft.md); the scripts live in code/ and write any file they produce beside themselves, which is why a fresh run leaves new

artefacts in code/ rather than overwriting the published results in the root.

Step 0 — environment. python3 -m pip install -r requirements-lock.txt. Output: nothing beyond pip's progress lines. Check: python3 docs/make_manifest.py --lock should print requirements-lock.txt OK.

Step 1 — archive script (Studies 1–3). python3 experiments.py. Writes experiment_results.json, fig_study1_convergence.png, fig_study2_robustness.png, fig_study3_precision.png and generated_tables.md. Measured wall clock: **196 s** (a second run on the same machine took 205 s). Key numbers in the output file: study2.T5_bartlett.skewed = 0.4288 (Bartlett under the skewed parent), study2.T9_median_test.small_freq = 0.0691 (the median test's worst rung), study1_ks_threshold = 0.01358, study3_nstar_mc ranging from 1500 (ANOVA) to 3000. The console ends with a five-seed stability check whose summary must contain, for example, Median test n=500: KS=0.0230±0.0034 — the value quoted in Section 5.1 of the manuscript.

Step 2 — sensitivity ladders and sample-size sweep. python3 code/rev_sensitivity.py. Writes rev_results.json, fig_rev_severity.png, fig_rev_study2_n.png. Measured wall clock: **70 s** (the file's own meta.runtime_seconds records the same figure). Key numbers: study2_nsweep.scenarios.skewed.tests.T5_bart = 0.4267, severity_sweep.ladders.skewness.tests.T5_bartlett.alpha_hat = [0.0525, 0.2665, 0.4323, 0.4323, 0.7717] along the five skewness rungs, and determinism_checks all true.

Step 3 — power companion. python3 code/rev_power.py. Writes rev_results_power.json and fig_rev_power.png. Measured wall clock: **53 s**. Key numbers: study4_power.verification.equivalence (the power path and the null path agree on the same data to within 2×10^{-13} for continuous statistics), and study4_power.verification.calibration (21/21 archived rates reproduced within 3σ). The command-line interface is:

```
usage: rev_power.py [-h] [--nmc NMC] [--no-fig] [--skip-verification]

Study 4 power companion (R1 comment 2)

options:
  -h, --help           show this help message and exit
  --nmc NMC            MC replications (default 10000)
  --no-fig             skip figure generation
  --skip-verification  skip the equivalence / calibration checks
```

Step 4 — 13-test Study 1, multi-α and null calibration. python3 code/rev_study1_ext.py (stages validate,study1,calib,t12diag,t13diag,figure). Writes rev_results_t1.json and fig_rev_alpha.png. Measured wall clock: **2302 s** (≈38 min; the recorded study-1/calibration block alone reports wall_clock_s = 1601). Key numbers: meta.core_selftest = "passed" and selftest 20/20 ok (the analytic guard rails for the Cramér–von Mises and Hoeffding reference laws, including the KPSS critical values); archive_reproduction_check.like_for_like_type1_max_abs_diff = 0.0, i.e. the $\alpha = 0.05$ column of T1–T10 is reproduced *exactly* from the archived RNG stream, with max_abs_diff_ks = 2.5\times10^{-14} (the floating-point reduction effect of Section 10); null_calibration for T11, T12 and T13; t12_reference_diagnostic, which shows that the finite-sample law matches an independently estimated limit law for $n \geq 50$ and that T12's large D_{KS} is a property of the published coarse table rather than of the statistic; and t13_split_diagnostic (grid_inflation_at_largest_n = 0.187, mc_vs_exact_at_largest_n = 0.0725), which attributes T13's liberality to the fixed flexible learner rather than to a numerical artefact. The verification suite alone is available without touching any results file:

```
$ python3 code/rev_study1_ext.py --validate
======================================================================
rev_study1_ext.py -- 14-test panel, multi-alpha, null calibration
```

```
N = 10000, ns = [5, 10, 20, 30, 50, 100, 200, 500], alphas = (0.01, 0.05, 0.1), quick = False
======================================================================

[1/6] verification suite
  [validate] selftest: 18 checks passed
  [validate] CvM limit CDF vs goftest: max |diff| = 4.43e-09
  [validate] Hoeffding D port vs canonical algorithm: max |diff| = 0.00e+00

--validate: verification suite complete; no results file written.

$ python3 -c "import core, json; s = core._selftest(); print(len(s), 'checks,', all(v['ok'] for v in s.v
18 checks, True
```

Step 5 — multi-α sweep and dependence. `python3 code/rev_alpha_dependence.py` (stages `alpha,dependence,figure`). Merges `alpha_sweep_13`, `alpha_sweep_13_crosscheck_archive`, `identity_checks`, `dependence_13` and `meta_alpha_dep` into `rev_results_t1.json`, and writes `fig_rev_dependence.png`. Measured wall clock: **288 s** (≈5 min). Key numbers: `identity_checks.max_abs_residual = 9.7\times10^{-17}` over the three tail-mass identities; `alpha_sweep_13_crosscheck_archive.max_abs_diff = 0.0070` against the archived $\alpha = 0.05$ column (expected, because the RNG stream differs — the block reports the comparison rather than asserting it); `dependence_13.ar1` for the t test and the sign test at $\rho \in \{0, 0.2, 0.5, 0.8\}$, $n \in \{20, 50, 100\}$, and `dependence_13.clustered` for ANOVA, Kruskal–Wallis and Cochran's Q.

Step 6 — sensitivity ladders and sample-size sweep. `python3 code/rev_sensitivity.py`. Writes `rev_results.json`, `fig_rev_severity.png`, `fig_rev_study2_n.png`. Measured wall clock: **76 s** (58–76 s across reruns; the file's own `meta.runtime_seconds` records the same figure). Key numbers: `study2_nsweep.scenarios.skewed.tests.T5_bartlett.by_n["50"].alpha_hat = 0.4267`, `severity_sweep.ladders.skewness.tests.T5_bartlett.alpha_hat = [0.0525, 0.2665, 0.4323, 0.4323, 0.7717]` along the five skewness rungs, and `determinism_checks` all `true`.

Step 7 — power companion. `python3 code/rev_power.py`. Writes `rev_results_power.json` and `fig_rev_power.png`. Measured wall clock: **53 s**. Key numbers: `study4_power.verification.equivalence` (the power path and the null path agree on the same data to within 2×10^{-13} for continuous statistics), and `study4_power.verification.calibration` (21/21 archived rates reproduced within 3σ). The command-line interface is:

```
usage: rev_power.py [-h] [--nmc NMC] [--no-fig] [--skip-verification]

Study 4 power companion (R1 comment 2)

options:
  -h, --help           show this help message and exit
  --nmc NMC            MC replications (default 10000)
  --no-fig             skip figure generation
  --skip-verification  skip the equivalence / calibration checks
```

Step 8 — independent power check. `python3 code/verify_new_tests_power.py`. Writes `rev_results_newtests_powe`
Measured wall clock: **37 s**. Key numbers: for T11 KPSS at $n = 100$ the size is $\hat{\alpha} = 0.0402$ and the nominal power 0.831; for T12 Hoeffding D at $n = 100$ the size is 0.0591 and the power against linear dependence 0.9973, both with Wilson intervals in the file.

Step 9 — verification. `python3 code/verify_manuscript.py` (235 checks, 0.3 s, see Section 10), `python3 docs/verify_bartlett.py` (13/13, 1 s) and `python3 docs/make_manifest.py --check` (whole-tree checksums).

Runtime summary. A complete regeneration of every number, table and figure in the paper takes about **50 minutes** on the machine of Section 3 (196 + 2302 + 288 + 76 + 53 + 37 s), dominated by the 13-test Study 1 grid with its null-calibration and diagnostic stages. A reader who only wants to check the package rather than regenerate it needs about two seconds (Steps 0 and 9). Detailed per-test timings are printed by each script as it runs; every run records its own wall clock in its `meta` block.

7. Function-level documentation

Each Python file documents its functions, so that the purpose of every routine, together with the format of its input and of its output, is readable without following the whole script. **Every top-level function of every script carries a docstring — 188 of 188 across the eight pipeline files** (`experiments.py` 20/20, `core.py` 44/44, `rev_study1_ext.py` 11/11, `rev_alpha_dependence.py` 7/7, `rev_sensitivity.py` 52/52, `rev_power.py` 38/38, `verify_new_tests_power.py` 3/3, `verify_manuscript.py` 3/3), and so does every tool in `docs/`. The archive script had a single documented function and was completed first; the three files that produce the sensitivity and power numbers were completed last.

The full list is generated from the sources by `docs/make_function_reference.py` and written to `docs/FUNCTION_REFERENCE` because it is generated, it cannot go stale. Library-style browsing works out of the box:

```
$ python3 -m pydoc core | head -10
Help on module core:

NAME core - core.py -- shared Monte Carlo kernel for the 14-test panel
(COST-D-26-00503R1).

DESCRIPTION This module is the long-lived home of the test panel used by
"Finite-Sample Adequacy and Robustness of Null Distributions". It is imported
by ``rev_study1_ext.py`` (t1), ``rev_alpha_dependence.py`` (t1),
``rev_sensitivity.py`` (t2) and ``rev_power.py`` (t3). ```

A representative example of the documentation style, from `core.py`:

```python
def _evaluate(stat_values, test_id, alpha=0.05, n=None, ref_args=None):
    """Evaluate a batch of statistics against a test's null reference.

    Args:
        stat_values: 1-D array of N Monte Carlo statistics.
        test_id: key of :data:`TESTS`, e.g. ``'T5_bartlett'``.
        alpha: nominal level at which the rejection rate is computed.
        n: per-group sample size, needed only when the reference law depends on n.

    Returns:
        dict with keys ``ks`` (D_KS), ``type1`` (alpha-hat), ``ci_lo``/``ci_hi``
        (95% Wilson), ``n_effective``, ``critical_value`` and ``alpha``.
    """
```

Every documentation pass was verified to be behavior-neutral, because these scripts produce the numbers in the paper. Two independent proofs were used, and both must hold:

- *The code is unchanged.* Parsing each file, stripping docstrings and hashing the resulting AST gives the same digest before and after the edit. For `experiments.py` that digest is
```

42c614978d1dc6e2dc4a6e2b0894eb696a028878617fcd7bbc5c507428794872

and for the seven frozen files (`core.py`, the four `rev_*.py`, the two `verify_*.py`) the seven digests `8ed4b2b9…`, `6458858b…`, `da82508d…`, `27726ed1…`, `f690a8d5…`, `2c81b2ce…`, `377f7219…` are likewise unchanged.

- *The behavior is unchanged.* `docs/probe_docstring_pass.py` imports all seven scripts, exercises a fixed set of functions with fixed seeds — seeded cells, the statistic constructors, the rejection rules, the three `core.py` severity hooks, the Wilson helpers, and the two read-only verification scripts as subprocesses — and hashes the results. The fingerprint

  d28d19dd0731caf6a7e344aef56d4679075c628d06e5acef9d4d754a09173f09

  is unchanged by spelling corrections to the prose, because the probe hashes the values its fixed-seed calls return rather than the text of the sources; the digest above is the current one, and it was identical before and after the documentation pass itself. The earlier `experiments.py` pass is covered by `docs/probe_experiments.py`, whose RNG digest 3824e5fe16564e764fec478cac090ffa7d6cd483e555c20c533d81 is likewise unchanged, and by a re-run of the whole archive script.

The verification scripts were re-run after the pass and are unaffected: `verify_manuscript.py` still reports `passed: 235  failed: 0`, and `rev_study1_ext.py --validate` still reports `core._selftest` 20/20.

8. Worked example: reproducing one published number

The manuscript reports that Bartlett's test rejects at rate $\hat{\alpha} = 0.429$ under the skewed parent Exp$(1) - 1$ with three groups of $n = 50$ and $N = 10{,}000$ replications (abstract; Table 5; Section 5.2). The stored machine value is 0.4288 in `experiment_results.json`. The command below re-runs the same configuration through the sensitivity pipeline and compares the printed number and its Wilson interval with the committed results file, character by character.

```
$ python3 docs/verify_bartlett.py
======================================================================
Worked example: Bartlett Type I error under the skewed parent Exp(1)-1
======================================================================
python 3.13.5   numpy 2.4.6   scipy 1.16.3
core.py seed=2024  N_MC=10000  alpha=0.05

[1] Exact reproduction through rev_sensitivity.py (committed seed)
    spec        : {'dist': ('exp', None), 'ns': [50, 50, 50], 'sd': [1.0, 1.0, 1.0]}
    cell seed   : 19204   (rev_sensitivity.cell_seed('nsweep', 'skewed'))
    N           : 10000
    alpha_hat   : 0.4267
    95% Wilson  : [0.417036, 0.436420]
   [PASS] rev_results.json  study2_nsweep ... by_n['50'].alpha_hat == printed value --
0.4267 == 0.4267  (exact float equality)
  [PASS] ...ci_lo matches the stored Wilson interval -- 0.417036026367
  [PASS] ...ci_hi matches the stored Wilson interval -- 0.436420267794
  [PASS] ...stored seed matches the seed used here -- 19204 == 19204

[2] Independent route through the shared kernel core.py (fresh seed)
   np.random.seed(7); core.mc_bartlett(3, 50, 10000, dist='expon'); core._evaluate(..., 'T5_bartlett'
    type1       : 0.4342   95% Wilson: [0.424513, 0.443938]
    KS distance : 0.4414   critical value chi2_0.95(2) = 5.991465
  [PASS] kernel run agrees with rev_results.json within 3 Monte Carlo sd -- z = +1.071
```

```
   archive generator alone: np.random.seed(2024); core.mc_bartlett(3, 50, 10000, dist='expon')
   type1     : 0.4328  -- NOT the archived 0.4288, because the archive stream is consumed by Study 1 fir
   [PASS] standalone archive call is within 3 Monte Carlo sd of the archived value --
z = +0.571

[3] Independent Monte Carlo realisations of the same scenario archive
experiment_results.json : study2.T5_bartlett.skewed = 0.4288 sensitivity
rev_results.json : study2_nsweep ... by_n['50'] = 0.4267 power
rev_results_power.json : verification.calibration.rows = 0.4329 kernel this
script (seed 7) = 0.4342 archive vs power : z = -0.585 (stored by
rev_power.py: +0.585) archive vs n-sweep: z = +0.300 [PASS] archive
realisation agrees with the power study's calibration check within 3 sd -- z =
-0.585 [PASS] archive realisation agrees with the sensitivity n-sweep within 3
sd -- z = +0.300 [PASS] rev_power.json stores the archive value and its
difference correctly -- archive_alpha_hat=0.4288, difference=0.0041 [PASS] the
manuscript's rounded value 0.429 equals round(0.4288, 3) -- round(0.4288, 3) =
0.429

[4] Closed-form constants used throughout the paper
    core.wilson_ci(0.05, 10000)      = (0.0458984827, 0.0544471158)
    half-width at alpha = 0.05       = 0.004274
    core.KS_THRESHOLD                = 0.0135800000
    1.358 / sqrt(10000)              = 0.0135800000
  [PASS] Wilson interval is centered on 0.05 and contains it -- [0.045898, 0.054447]
   [PASS] Wilson half-width is about 0.004 (manuscript says approx. 0.004) -- half-
width = 0.004274
  [PASS] KS threshold equals 1.358/sqrt(N) to machine precision -- np.float64(0.013580000000000002)

======================================================================
13/13 checks passed
Worked example reproduced: Bartlett alpha-hat = 0.4267 (rev_results.json, n = 50, N = 10,000, skewed par
consistent with the 0.4288 archive value reported as 0.429 in the paper.
======================================================================
```

Four things are worth reading out of this output.

- **The pipeline reproduces its own stored number exactly.** Part [1] re-runs the cell with the seed that produced `rev_results.json` and finds $\hat{\alpha} = 0.4267$ with Wilson interval $[0.417036, 0.436420]$, identical to the stored values to the last digit.
- **An independent route gives the same answer.** Part [2] goes through the shared kernel with a different seed and obtains 0.4342, which differs from the stored value by $z = 1.07$ — well inside Monte Carlo error. The difference between two independent runs is *supposed* to be non-zero; what must be reproducible is the value given the seed, and the size of the discrepancy given the same design.
- **A standalone generator call gives a different, equally valid number.** The extra lines in part [2] show 0.4328 for the same configuration when the generator is called on its own after `np.random.seed(2024)`. The archived value 0.4288 comes from a later point of the same global stream (the archive script has already consumed it for Study 1), and the two differ by $z = 0.57$ — inside Monte Carlo noise. Section 8.1 explains the mechanism.
- **Three independent realisations agree.** The archive run (0.4288), the sensitivity n-sweep (0.4267) and the power study's calibration check (0.4329) are three different random samples of the same configuration; all pairwise differences are below one standard error, and the paper rounds the archive value to 0.429.

- **The two constants quoted in the manuscript are exact.** The KS adequacy threshold $1.358/\sqrt{10{,}000} = 0.01358$ and the Wilson half-width near $\hat{\alpha} = 0.05$ (0.0043, quoted as "approximately 0.004") are computed, not assumed.

8.1 Why a single number can differ between runs

A reader comparing their own run with the committed files will sometimes find a value that is close but not identical. Three distinct mechanisms explain every such case, and none of them is an error.

1. **The archive script has one global random stream.** `experiments.py` calls `np.random.seed(2024)` once, at import, and every generator then draws from that single stream. A stored value is therefore reproducible only by running the whole script in the original order: the state of the stream when Bartlett's Study 2 cell is drawn depends on everything drawn before it. Calling `mc_bartlett(3, 50, 10000, dist='expon')` on its own after `np.random.seed(2024)` gives 0.4328 rather than the stored 0.4288 — an equally valid realisation of the same configuration, drawn from a different point of the stream.
2. **The sensitivity and power scripts take an explicit seed per cell.** `rev_sensitivity.py` derives a stable seed from the cell's identity (`cell_seed("nsweep", "skewed")` gives 19204), and `core.py` uses `np.random.default_rng(seed)` throughout, so any single configuration can be reproduced on its own without replaying anything else. Two runs of the same cell return bit-identical arrays; this is the property the worked example exercises in part [1].
3. **Two independent Monte Carlo realisations of the same configuration differ by sampling noise.** At $\alpha = 0.05$ and $N = 10{,}000$ the standard error of a rate is $\sqrt{\alpha(1-\alpha)/N} \approx 0.0022$, so differences of a few thousandths between runs are expected. Bartlett under skewness is 0.4288 in the archive run, 0.4267 in the sensitivity n-sweep and 0.4329 in the power study; the differences are $z = 0.30$ and $z = 0.59$, well inside noise, and all three round to 0.43 at the precision reported in the paper.

Regeneration rule. A reported value is reproducible bit-for-bit by re-running the script that produced it with the stated seed; standalone calls to individual generators are reproducible only under the RNG-state conventions documented in `docs/CODE_MAP.md` §6. Where a comparison matters, use the Wilson intervals that every rate carries rather than the raw digits: two rates whose intervals overlap are not distinguishable at that Monte Carlo budget.

9. Guidance for readers who do not use Python

Nothing in this package requires programming experience beyond editing one number in a text file. The steps below are the shortest path; screenshots are omitted, but every step is standard.

Step 1 — install Anaconda (once). Download *Anaconda Distribution* from https://www.anaconda.com/download for macOS, Windows or Linux and run the installer with the default options. Anaconda provides Python together with the scientific libraries used here, and installs a graphical program called **Spyder** that behaves like a statistics package: an editor on the left, a variable explorer, and an output console.

Step 2 — get the code. Download the repository from its URL (Section 11) with the green *Code* button, then *Download ZIP*, and unpack it, or, if you use GitHub Desktop, clone it. You should end up with a folder containing `experiments.py`, `core.py`, `rev_sensitivity.py`, `rev_power.py` and a `docs` folder.

Step 3 — install the exact package versions (once). Open the *Anaconda Prompt* (Windows) or *Terminal* (macOS/Linux), move into the folder you just unpacked, and run:

```
cd path/to/Finite-Sample\ Adequacy\ and\ Robustness\ of\ Null\ Distributions
python3 -m pip install -r requirements-lock.txt
```

This downloads the four libraries used by the study and their dependencies. It takes a few minutes and needs an internet connection. If `python3` is not found, open the program *Anaconda Prompt* instead of the system terminal — Anaconda sets the command up for you.

Step 4 — check that everything works (one second). In the same window run:

```
python3 docs/verify_bartlett.py
```

You should see `13/13 checks passed` at the end. This re-runs one configuration from the paper and compares it with the published results file, so a green result means your environment matches the paper's environment.

Step 5 — run a study. To regenerate the main results:

```
python3 experiments.py
```

The script prints its progress test by test and finishes after about three and a half minutes with `Results saved to experiment_results.json`. It writes its output into `results/`, so if you want to keep the published files, copy the folder first.

Changing a parameter. All parameters sit at the top of the script that uses them, in plain sight, and each is documented in the docstring above its function. Three examples:

- *Change the number of replications.* In `rev_power.py` either edit the default `--nmc 10000` in `main()` or simply pass it on the command line: `python3 code/rev_power.py --nmc 2000` (a fifth of the work, about ten seconds).
- *Change the sample size.* In `experiments.py`, `run_study1` contains `sample_sizes = [5, 10, 20, 30, 50, 100, 200, 500]`; replace it with, say, `[50, 100]` to study only two sizes. In `rev_sensitivity.py`, the sweep grid is `NSWEEP_N` at the top of the file.
- *Change the nominal level.* Both `experiments.py` and the sensitivity scripts take α from a small set; in `core.py` it is `ALPHAS = (0.01, 0.05, 0.10)`. Add a value there and every study that calls `evaluate_multi_alpha` will report it.

Reading the output. Every results file is JSON: plain text that opens in any editor, with one entry per test and scenario. For example, `experiment_results.json` contains `"T5_bartlett": {"baseline": 0.0479, "skewed": 0.4288, ...}`, and every sensitivity result file additionally starts with a `meta` block recording the seed, the package versions and the timestamp of the run.

If something goes wrong. Three common cases and their fixes:

- `ModuleNotFoundError: No module named 'numpy'` — the packages were installed into a different Python. Run `python3 -m pip install -r requirements-lock.txt` using the same command name (`python3`) that you use to run the scripts, or use the Anaconda Prompt.
- `ModuleNotFoundError: No module named 'experiments'` — the command was run from the wrong folder. The scripts import each other by file name, so the working directory must be the folder that contains `experiments.py`.
- The figures come out on screen instead of being saved, or the run stalls after the last test — on a machine without a display, set the environment variable `MPLBACKEND=Agg` before running. On macOS and Linux this is `MPLBACKEND=Agg python3 experiments.py`; the sensitivity scripts already do this internally.

Getting help. Please open an issue in the repository (Section 11) rather than writing to the authors directly: the answer then helps the next reader.

10. Verifying that a run succeeded

Six independent checks are available; all of them were executed for this vignette, with the output recorded in `docs/verification_log.txt`.

(1) The environment matches the lock file.

```
$ python3 docs/make_manifest.py --lock
lock entries checked : 13
```

```
requirements-lock.txt OK
```

(2) The files match the checksum manifest. `docs/MANIFEST.sha256` lists the SHA-256 of every pipeline script, results file, figure, build file and `docs/` tool (38 files). Regenerate it after a run with `python3 docs/make_manifest.py`, or check a downloaded copy with `--check`:

```
$ python3 docs/make_manifest.py --check
manifest entries : 38
missing          : none
changed          : none
unlisted new     : none
MANIFEST OK
```

The manifest covers every Python source in `code/`, every results JSON and published figure, `build.sh` and `latex_preamble.tex` in the package root, plus the `docs/` tools, the two environment records and the rendered flowchart. Excluded on purpose, and stated in the manifest header: the manifest itself, the verification log (rewritten by every run) and the Markdown documents; the reduced-budget smoke artefact `rev_results_t1_quick.json` is also excluded.

(3) The worked example reproduces a published number — Section 8; `13/13 checks passed`, exit status 0.

(4) The results files carry their own provenance. Every sensitivity results file starts with a `meta` block written at run time. For `rev_results.json`:


```
{"seed": 2024, "N_mc": 10000, "n_fixed": 50,
 "alpha_levels": [0.01, 0.05, 0.1],
 "numpy": "2.4.6", "scipy": "1.16.3", "python": "3.13.5",
 "git_hash": "unknown",
 "timestamp": "2026-10-12T22:26:04+0800",
 "script": "rev_sensitivity.py", "script_sha256": "8a4fcc364b128ac7",
 "n_cells_ladders": 156, "n_cells_nsweep": 144,
 "unique_cached_draws": 343, "runtime_seconds": 75.8,
 "figures_written": true}
```


`git_hash` is `unknown` because the working copy is not a git checkout; in the released repository it records the commit that produced the file. Because the archive script writes plain JSON without a `meta` block, its provenance is given by the run itself (the console prints the seed, the grid and the timings) and by the checksum manifest.

(5) Every number in the manuscript is backed by a results file. `verify_manuscript.py` is deliberately adversarial: it does not trust `draft.md`. Each check names the manuscript claim and the JSON path that should produce it, and compares them at the tolerance implied by the Monte Carlo budget; any mismatch prints `FAIL` and sets the exit code to 1. It is the check to re-run after editing either the manuscript or a results file:

```
$ python3 code/verify_manuscript.py
[1] Archived Study 2 cells (experiment_results.json -> Table 5)
[2] Archived Study 1 / Study 3 (Table 4, Table 7)
[3] Severity ladders (rev_results.json -> Section 5.4)
[4] Sample-size sweep (rev_results.json -> Table 10)
[5] Alpha sweep (rev_results.json -> Section 5.4 S1)
[6] Power study (rev_results_power.json -> Tables 11-12)
[7] Source hygiene
  tables: 15 captions, refs to [1, 2, 3, 4, 5, 7, 8, 9, 10, 11, 13, 14, 15]
[8] New-test panel (rev_results_t1.json -> Tables 4-6, 12)
[9] New-tests power (rev_results_newtests_power.json)
```

```
[10] Text claims that must be consistent with the JSON

passed: 235   failed: 0   skipped: 0
ALL CHECKS PASSED
```

It runs in 0.3 s and needs nothing beyond the standard library.

(6) A clean environment built only from the lock file is sufficient. As the strongest test that `requirements-lock.txt` is complete, we built a fresh virtual environment containing nothing but those thirteen packages and ran the worked example in it:

```
$ python3 -m venv /tmp/mctk_venv
$ /tmp/mctk_venv/bin/python -m pip install -r requirements-lock.txt
Successfully installed contourpy-1.3.3 cycler-0.12.1
  fonttools-4.63.0 kiwisolver-1.5.0 lightgbm-4.6.0
  matplotlib-3.11.0 numpy-2.4.6 packaging-26.2 pillow-12.3.0
  pyparsing-3.3.2 python-dateutil-2.9.0.post0 scipy-1.16.3 six-1.17.0
$ /tmp/mctk_venv/bin/python docs/verify_bartlett.py
...
13/13 checks passed
Worked example reproduced: Bartlett alpha-hat = 0.4267
(rev_results.json, n = 50, N = 10,000, skewed parent),
consistent with the 0.4288 archive value reported as 0.429
in the paper.
```

No package outside the lock file was needed.

Key numbers per file, for a reader comparing their own run with the committed one:

File	Key path	Value
`experiment_results.json`	`study2 > T5_bartlett > skewed`	0.4288
	`study2 > T9_median_test > small_freq`	0.0691
	`study1_ks_threshold`	0.01358
	`study3_nstar_mc > T2_anova`	1500
`rev_results.json`	`study2_nsweep > scenarios > skewed > tests > T5_bartlett > by_n["50"] > alpha_hat`	0.4267
	`severity_sweep > ladders > skewness > tests > T5_bartlett > alpha_hat[2]`	0.4323
`rev_results_power.json`	`study4_power > verification > calibration > all_consistent`	`true`
	`study4_power > verification > calibration > rows[10] > alpha_hat`	0.4329
`rev_results_t1.json`	`meta > core_selftest`	`passed`
	`archive_reproduction_check > like_for_like_type1_max_abs_diff`	0.0
	`identity_checks > max_abs_residual`	9.7e-17

File	Key path	Value
rev_results_newtests_power.json	t13_split_diagnostic > grid_inflation_at_largest_n	0.187
	T11_kpss > n=100 > power_nominal	0.831
	T12_hoeffding_d > n=100 > power_linear	0.9973

How exactly does a rerun reproduce the committed files? We ran the archive script twice end to end on the machine of Section 3 (196 s and 205 s). The two runs produced **byte-identical** results files (SHA-256 `7fb7080f…`), which shows the script is deterministic within a fixed environment. Compared with the committed result file (SHA-256 `6d533636…`), all integer rejection rates, all n^* values and all intervals agree exactly; 29 of the ~1,000 Study 1 floating-point KS and KL distances differ in the 14th significant digit (largest relative difference 2.9×10^{-12}), which is floating-point reduction order in a different numerical environment. No value at the precision reported in the paper (three decimals) changes. This is the honest form of the reproducibility claim: the pipeline is deterministic, and a small part of the archive output is reproducible to about twelve significant digits rather than bit for bit when the numerical libraries are not the exact versions pinned here.

The same effect shows up in `rev_results_t1.json`, which re-generates T1–T10 from the archived RNG stream inside the sensitivity pipeline and compares them with `experiment_results.json`: `archive_reproduction_check` records `like_for_like_type1_max_abs_diff = 0.0` (the rejection counts match the archived denominator exactly) and `max_abs_diff_ks = 2.5\times10^{-14}`, the same floating-point reduction effect, found independently by a different script.

11. Repository, license and data availability

The GitHub link in the manuscript points at a **dedicated project folder inside the mc-tk repository that contains only this study's material**:

`https://github.com/zhangys11/mc/tree/main/Finite-Sample%20Adequacy%20and%20Robustness%20of%20Null%2`

which holds the scripts, results, figures, this vignette and the documentation, and nothing else; the same folder is included as Supplementary Material so the materials remain available independently of the repository. The toolkit repository itself (`https://github.com/zhangys11/mc`) is cited only as the *software* reference for Section 4. `docs/REPO_LAYOUT.md` documents the layout, the licensing (MIT for code, CC BY 4.0 for results and figures), the alternative of a separate repository with a Zenodo DOI, and the exact wording proposed for the Data Availability statement and for the point-by-point response. The Data Availability statement should point at the project folder above rather than at the repository root, because the root holds unrelated work and does not identify the package unambiguously.

Appendix A. How this vignette was built

The Markdown source is `vignette.md`; the PDF was produced with pandoc and XeLaTeX:

```
pandoc vignette.md -o vignette.pdf \
  --pdf-engine=xelatex --toc \
  -V geometry:margin=1.9cm -V fontsize=10pt \
  -V mainfont="Helvetica Neue" -V monofont="Menlo" \
  --highlight-style=tango
```

The flowchart panel `docs/fig_execution_flow.png` was rendered by `python3 docs/make_flowchart.py`, the function reference by `python3 docs/make_function_reference.py`, the lock files by `python3 docs/make_lock.py`, and the manifest by `python3 docs/make_manifest.py`. `docs/run_verification.sh`

executes every command quoted in this vignette and writes `docs/verification_log.txt`; that log is the evidence that all of them completed successfully.

Appendix B. Files in this package

The package has two layouts, and the table gives the paths used in the public repository copy. In the submission copy the results sit at the package root instead of in `results/`, the analysis scripts sit beside them instead of in `code/`, and the manuscript, the response letter and the cover letter are present; those three are submission material and are not distributed here.

Path	Contents
`vignette.md`, `vignette.pdf`	this document
`README.md`	landing page: what the package can and cannot regenerate, and the cost of each
`requirements.txt`, `requirements-lock.txt`	direct dependencies; exact pins
`supplementary_material.md`	supplementary material, Sections S1–S10
`docs/CODE_MAP.md`	file-by-file purpose, inputs, outputs, runtime, dependencies
`docs/FUNCTION_REFERENCE.md`	generated function reference
`docs/REPO_LAYOUT.md`	repository proposal, licensing, Data Availability wording
`docs/ENVIRONMENT.txt`, `docs/pip_freeze_full.txt`	environment record
`docs/MANIFEST.sha256`	SHA-256 of every script, result and figure
`code/verify_manuscript.py`, `code/verify_crossrefs.py`, `code/verify_letter_claims.py`, `code/verify_new_tests_power.py`, `code/verify_dcor_against_R.py`	verification and independent cross-checks
`docs/verify_bartlett.py`, `docs/probe_experiments.py`, `docs/probe_docstring_pass.py`	worked example, determinism probe and behavioral fingerprint
`docs/make_lock.py`, `docs/make_manifest.py`, `docs/make_flowchart.py`, `docs/make_function_reference.py`, `docs/run_verification.sh`, `docs/regenerate.sh`, `docs/verification_log.txt`	tooling and its log
`docs/flowchart.mmd`, `docs/fig_execution_flow.png`, `docs/fig_execution_flow.pdf`	flowchart source and rendering
`build.sh`, `latex_preamble.tex`	manuscript, response-letter and supplement build (submission copy)
`experiments.py`	frozen original-submission script and driver
`code/core.py`, `code/rev_study1_ext.py`, `code/rev_alpha_dependence.py`, `code/rev_sensitivity.py`, `code/rev_power.py`, `code/rev_dcor.py`, `code/rev_imbal_hetero.py`, `code/archive_experiments.py`	the simulation pipeline
`code/gen_t12_law.py`, `code/gen_dcor_local.py`, `code/gen_t14_study3.py`	generators for the three long runs whose results are committed rather than re-derived
`code/make_fig2.py`, `code/postprocess_docx.py`	figure regeneration, and Word post-processing
`results/`	every results file, one per script, each with its seed and wall-clock time
`figures/`	the manuscript's panels

Path	Contents
`experiment_results.json`, `generated_tables.md`, `fig_study1/2/3_*.png`	frozen original-submission results, tables and figures